

We don't need addition for the Leibniz rule

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*Preliminary version**

Abstract

We show that for a linear differentiation, the Leibniz rule is equivalent to the condition that all its exponentials lie on a single ray in a vector space. We generalize the notions of a multiplicative exponential and a derivative to right rings and establish conditions under which the kernel of a differentiation embeds into the constants. We prove an analogue of the homomorphism theorem for differentiations and give an explicit form of certain differentiations of a triangular ring that lie beyond the scope of the known classification.

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Introduction

Gottfried Wilhelm Leibniz discovered his famous rule thanks to a mistake. Initially, he assumed the multiplicativity of the derivative. However, he quickly refuted this assumption geometrically and, after some time of reflection, arrived at the formula:

$$d(uv) = u d(v) + v d(u) + d(u) d(v)$$

Leibniz simply discarded the term $d(u)d(v)$, considering it “negligible,” although the knowledge of his time did not allow him to justify this step rigorously. A great mathematical discovery came into being thanks to an unforgivable carelessness — and this is how calculus was being born. The present paper also came into being thanks to a mistake — we discovered an error in the paper “On Algebraic Integral” [1], and all subsequent reasoning was initially motivated by the desire to correct it.

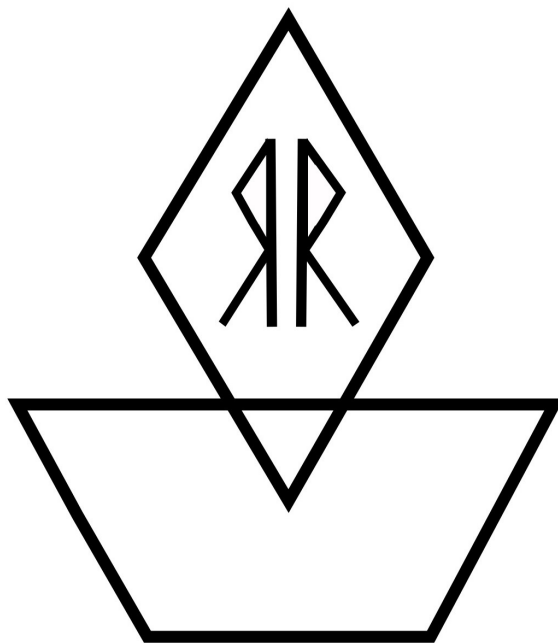
In turn, the paper “On Algebraic Integral” was written as the result of many years of reflection on what everyday intuition lies behind the chain rule — mathematicians call such intuition algebra, and ultimately we managed to find it. In that paper, we showed that the chain rule is closely connected with the notion of distributivity, which allowed us to give a purely algebraic definition of the integral without using such concepts as smoothness, continuity, or limits in any form. In it, as well as in the paper “On Algebraic Integral for Dummies” [2], we showed that the multiplicativity of the exponential is equivalent to the Leibniz rule.

In this paper, we continue to develop algebraic calculus and do for the Leibniz rule what we did for the chain rule in [1] — we investigate what everyday intuition lies behind the Leibniz rule. In Theorem 2, we show that a linear differentiation satisfies the Leibniz rule if and only if all its exponentials lie on a single ray in a vector space — this intuition is simultaneously algebraic and geometric, as is so characteristic of linear algebra.

In the process of writing this paper, we also repeatedly made mistakes and rewrote it — at the beginning of its writing, we assumed that any linear differentiation has a multiplicative exponential. However, linearity alone was not enough, and in the end it turned out that linearity has no essential bearing on the matter, and we were able to generalize the multiplicativity of the exponential and the notion of a derivative to right rings — a structure in which there is no operation of addition at all — and Lemma 3.2 “On the

embedding” is the main result of this paper. Theorem 2, in turn, proved for linear differentiations, gives an equivalent formulation of the Leibniz rule that does not use the operation of addition as well.

In turn, in Lemma 2.6 “On the kernel of a differentiation,” we demonstrate how algebraic calculus borrows methods from group theory, thereby allowing the proof of nontrivial results later on. In Section 5, we discuss an existing attempt to classify differentiations over C^1 and find an explicit form for certain differentiations in a triangular rring — including ones that lie beyond the scope of the known classification. In the conclusion, we demonstrate the beauty and nontriviality of certain differentiations, and these are undoubtedly only the first steps in the study of the astonishing and beautiful world of algebraic calculus.



1 Preliminaries

Definition 1.1. $\langle R, \cdot, \circ \rangle$ is called a **right ring** (or **rring**) if it is a near-ring with unity id where $\langle R, \cdot \rangle$ is a commutative group. That is, a ring with unity that does not necessarily satisfy left distributivity. For notational convenience, we will assume that composition $\circ$ takes precedence over multiplication $\cdot$.

Definition 1.2. Let $\langle R, \cdot, \circ \rangle$ be a right ring. An element $c \in R$ is called a **constant** if for all $f \in R$, $c \cdot f = c$.

Definition 1.3. $\langle R, \cdot, \circ \rangle$ is called a **division right ring** (or **division rring**) if it is a right ring and every nonconstant element $f \in R$ has a multiplicative inverse f^{-1} such that: $f^{-1} \cdot f = f \cdot f^{-1} = \text{id}$

Definition 1.4. $\langle R, +, \cdot, \circ \rangle$ is called a **triangular right ring** (or **triangular rring**) if:

1. $\langle R, +, \cdot \rangle$ is a field with sum $+$, multiplication $\cdot$, additive inverse $f \rightarrow -f$, multiplication inverse $f \rightarrow 1/f$, zero 0 and unit 1 ,
2. $\langle R \setminus \{0\}, \cdot, \circ \rangle$ is a division right ring with multiplication $\cdot$, composition $\circ$, additive inverse $f \rightarrow 1/f$, multiplication inverse $f \rightarrow f^{-1}$, zero $1 \neq 0$ and nonconstant unit $\text{id} \neq 1$.
3. Composition is right distributive over addition, that is:

$$\forall a, b, c \in R : (a + b) \circ c = a \circ c + b \circ c$$

Definition 1.5. We call a map $d : R \rightarrow R$ a **differentiation** of a right ring R if:

$$\forall f, g \in R : d(f \circ g) = d(f) \circ g + f \cdot d(g)$$

2 Constants and the kernel of a differentiation

In Lemma 3.10 of [1], it was shown that the constants form a two-sided ideal in $\langle R, \cdot, \circ \rangle$. Let us show that they also form a subfield in $\langle R, +, \cdot \rangle$.

Lemma 2.1. Let $\langle R, +, \cdot, \circ \rangle$ be a triangular ring. The constants form a subfield in $\langle R, +, \cdot \rangle$.

Proof. Let C be the set of constants of R . Let $c, t \in C, f \in R$.

Zero is a constant:

$$0 \circ f = (0 + 0) \circ f = 0 \circ f + 0 \circ f$$

$$0 = 0 \circ f$$

Additive inverses are constants:

$$0 = (c - c) \circ f = c \circ f + (-c) \circ f$$

$$(-c) \circ f = -(c \circ f) = -c$$

Constants closed under multiplication:

$$(c + t) \circ f = c \circ f + t \circ f = c + t$$

Thus, the constants form an additive subgroup in $\langle R, + \rangle$. The fact that the constants form a multiplicative subgroup is proved similarly, using the right distributivity of composition over multiplication. $\square$

It is a well-known fact from algebra that a field forms a linear space over any of its subfields. Thus, a triangular ring forms a linear space over the ideal of its constants.

Definition 2.2. Let $\langle R, +, \cdot, \circ \rangle$ be a triangular ring. Let C be the ideal of its constants. We call a differentiation $d : R \rightarrow R$ **linear** if it is linear with respect to addition, that is,

$$\forall f, g \in R, \forall \alpha, \beta \in C : \quad d(\alpha f + \beta g) = \alpha d(f) + \beta d(g).$$

Definition 2.3. Let $\langle R, \cdot, \circ \rangle$ be a rring and let $d : R \rightarrow R$ be a differentiation. We call the set

$$\ker(d) = \{f \in R \mid d(f) = 1\}$$

the **kernel** of the differentiation d .

Lemma 2.4. If a differentiation equals zero on an invertible element, then it is identically zero. Let $d(f) = 0$ for some nonconstant $f \in R$:

$$d(x) = d(f^{-1} \circ f) = d(f^{-1}) \circ f \cdot d(f) = 0$$

So, for any $g \in R$:

$$d(g) = d(x \circ g) = d(x) \circ g \cdot d(g) = 0$$

- since zero is a constant by Lemma 2.1 $\square$

In what follows, we assume that all differentiations are not identically zero.

Lemma 2.5. It is easy to see that the kernel of a differentiation forms a compositional subgroup.

Proof. Let $\langle R, \cdot, \circ \rangle$ be a rring and let d be a differentiation. Let $f, g \in \ker(d)$.

id in $\ker(d)$:

$$d(\text{id}) = d(\text{id} \circ \text{id}) = d(\text{id}) \circ \text{id} \cdot d(\text{id}) = d(\text{id}) \cdot d(\text{id})$$

$$1 = d(\text{id})$$

Compositional inverses are in $\ker(d)$, since 1 is a constant:

$$1 = d(\text{id}) = d(f \circ f^{-1}) = d(f) \circ f^{-1} \cdot d(f^{-1}) = d(f^{-1})$$

$\ker(d)$ closed under multiplication:

$$d(f \circ g) = d(f) \circ g \cdot d(g) = 1 \quad \square$$

Lemma 2.6. On the kernel of a differentiation.

Let $\langle R, \cdot, \circ \rangle$ be a rring and let $d : R \rightarrow R$ be a differentiation. Then for any $f, g \in R$,

$$d(f) = d(g) \iff \exists k \in \ker(d) : f = k \circ g$$

Proof. Let $d(f) = d(g)$ for some $f, g \in R$, then:

$$\begin{aligned} d(f \circ g^{-1}) &= d(f) \circ g^{-1} \cdot d(g^{-1}) = d(g) \circ g^{-1} \cdot d(g^{-1}) \\ &= d(g \circ g^{-1}) = d(\text{id}) = 1, \end{aligned}$$

so, $f = (f \circ g^{-1}) \circ g$ and $f \circ g^{-1} \in \ker(d)$

Conversely, let $f = k \circ g$ for some $k \in \ker(d)$,

$$d(f) = d(k \circ g) = d(k) \circ g \cdot d(g) = (1 \circ g) \cdot d(g) = d(g) \quad \square$$

Remark. The statement of Lemma 2.6 is equivalent to the homomorphism theorem from group theory. However, the composition group R cannot always be factored by the kernel, since, unlike the kernel of a homomorphism, the kernel of a differentiation is not necessarily normal:

$$d(f \circ k \circ f^{-1}) = d(f) \circ k \circ f^{-1} \cdot d(k \circ f^{-1}) = d(f) \circ k \circ f^{-1} \cdot d(f^{-1})$$

So, $f \circ k \circ f^{-1} \notin \ker(d)$ in general.

Definition 2.7. We call the kernel of a differentiation **trivial** if $\ker(d) = \{\text{id}\}$.

Corollary of Lemma 2.6. The kernel of a differentiation is trivial **if and only if** the differentiation is injective.

Proof. If $\ker(d)$ is nontrivial, then there exists $f \neq \text{id}$ with $d(f) = 1 = d(\text{id})$, so d is not injective. Conversely, suppose d is not injective, so there exist $f, g \in R$ with $f \neq g$ and $d(f) = d(g)$. By the lemma, $d(f \circ g^{-1}) = 1$, but $f \neq g$, so we have $f \circ g^{-1} \neq \text{id}$, and $\ker(d)$ is nontrivial. $\square$

Definition 2.8. Let $\langle R, \cdot, \circ \rangle$ be a rring and let $d : R \rightarrow R$ be a differentiation. An element $e \in R$ is called an **exponential element of d** if it is a fixed point of f , i.e., $d(e) = e$ and e is nonconstant. We call the compositional inverse of the exponential element the logarithm: $\ln \circ e = e \circ \ln = \text{id}$

Remark. As for constant exponentials, the situation is very simple. The only such exponentials of a differentiation can be 1 or 0. Let c be a constant exponential of d (not equal to 1 or 0), that is, $d(c) = c$. Then:

$$c = d(c) = d(c \circ c) = d(c) \circ c \cdot c = c \cdot c$$

In particular, 0 and 1 are the only constant operators that are differentiations. On the other hand, if a differentiation maps some constant elements to 1, or, in other words, if the kernel of a differentiation contains constants, then the differentiation is identically equal to 1. Let $d(c) = 1$ and $f \in R$:

$$1 = d(c) = d(c \circ f) = d(c) \circ f \cdot d(f) = d(f)$$

Thus, the only constant exponential of a nontrivial differentiation can be zero.

3 Theorem 1. On Bijection

Let $\langle R, \cdot, \circ \rangle$ be a right ring and let d be a differentiation of R with an exponential e . Let E be the set of all exponentials of d . Then there exists a bijection $\varphi : \ker(d) \rightarrow E$, given by:

$$\varphi(k) = e \circ k$$

Proof. We first verify that φ indeed maps elements of the kernel $\ker(d)$ to the set of exponentials E . Let $k \in \ker(d)$, then:

$$d(\varphi(k)) = d(e \circ k) = d(e) \circ k \cdot d(k) = e \circ k$$

- so $\varphi(k)$ is indeed an exponential of d . It is easy to verify that the inverse of φ is the map $\psi : E \rightarrow \ker(d)$ given by $\psi(e') = \ln \circ e'$, where $\ln$ is the logarithm of e :

$$\psi(\varphi(k)) = \ln \circ (e \circ k) = k, \quad \varphi(\psi(e')) = e \circ (\ln \circ e') = e'$$

To verify that φ is surjective, we show that ψ indeed maps exponential elements into the kernel. Let e' is some exponential of d . By Theorem 4 from [1], it follows that $d(\ln) = \frac{1}{\text{id}}$, so:

$$d(\psi(e')) = d(\ln \circ e') = d(\ln) \circ e' \cdot d(e') = \frac{1}{e'} \cdot e' = 1$$

Therefore, φ and ψ are bijections, and $|\ker(d)| = |E|$. ■

Remark. $\frac{1}{\text{id}} \circ f = \frac{1}{f}$ follows from the fact that 1 is a constant and that composition is distributive over multiplication:

$$(\frac{1}{\text{id}} \cdot \text{id}) \circ f = 1$$

Remark. The theorem does not cover constant exponentials, since its proof uses the logarithm, which a constant exponential obviously cannot have.

Remark. The bijection φ is non-canonical, as it depends on the choice of the fixed exponential e . For any other exponential $\tilde{e} = e \circ h$ with $h \in \ker(d)$, the corresponding bijection $\tilde{\varphi}$ satisfies $\tilde{\varphi}(f) = \varphi(h \circ f)$. Thus, all such bijections differ from each other by composition with elements of the kernel group.

Definition 3.1. Let $\langle R, \cdot, \circ \rangle$ be a right ring and let C be the ideal of its constants. We call a differentiation d of R a **derivative** if its set of exponentials E is nonempty and for any $e_1, e_2 \in E$, $e_1/e_2 \in C$.

Lemma 3.2. On the embedding

Let $\langle R, \cdot, \circ \rangle$ be a right ring, let d be a **derivative** on R , and let E be the set of exponentials of d . Suppose there exists $e \in E$ such that $e \circ 0 = 1$. Then the composition group of the **kernel** embeds **into** the multiplicative group of the **constants** via an injective homomorphism $\theta : \langle \ker(d), \circ \rangle \hookrightarrow \langle C, \cdot \rangle$ given by $\theta(k) = e \circ k \circ 0$.

Proof.

Define the function $b : \ker(d) \rightarrow C$ by

$$(1) \quad b(k) = \frac{e \circ k}{e}$$

On the one hand, the bijection theorem shows that $e \circ k$ is an exponential for any $k \in \ker(d)$. On the other hand, d is a derivative, so the quotient of any two of its exponentials is a constant. Hence, b indeed maps the kernel into C .

Also, it follows from the bijection theorem that b is **injective**. By the theorem, the exponential e defines a bijection φ between the kernel and the set of exponentials. Therefore, for all $k_1, k_2 \in \ker(d)$,

$$b(k_1) = b(k_2) \iff \frac{e \circ k_1}{e} = \frac{e \circ k_2}{e} \iff \frac{\varphi(k_1)}{e} = \frac{\varphi(k_2)}{e}$$

so $k_1 = k_2$ by the injectivity of φ .

We now show that b is a **multiplicative homomorphism**. It is easy to see that, by reasoning analogous to Lemma 2.1, the constants form a multiplicative subgroup in a right ring, just as they form a subfield in a triangular ring. By the definition of b , for all k in $\ker(d)$:

$$(2) \quad e \circ k = b(k)e$$

Let $k, t \in \ker(d)$. Since the kernel is a composition group, and by (2),

$$(3) \quad e \circ k \circ t = b(k \circ t)e$$

Now compositionally multiply (2) on the right by t :

$$e \circ k \circ t = (b(k)e) \circ t = (b(k) \circ t)(e \circ t)$$

— by distributivity, and since $b(k)$ is a constant,

$$= b(k)(e \circ t)$$

But $e \circ t = b(t)e$ by (2). Therefore,

$$e \circ k \circ t = b(k)b(t)e$$

Equating this with (3),

$$b(k \circ t)e = b(k)b(t)e$$

and dividing by e , we obtain the multiplicativity of b :

$$b(k \circ t) = b(k)b(t)$$

It **remains to show** that $\theta(k) = b(k)$ for any $k \in \ker(d)$. To this end, compositionally multiply (1) on the right by 0:

$$b(k) \circ 0 = \left(\frac{e \circ k}{e} \right) \circ 0 = \frac{e \circ k \circ 0}{e \circ 0}$$

Since $b(k)$ is a constant, and $e \circ 0 = 1$ by the hypothesis of the lemma,

$$b(k) = e \circ k \circ 0$$

Thus,

$$\begin{aligned} \theta(k \circ t) &= e \circ k \circ t \circ 0 = b(k \circ t) = b(k)b(t) \\ &= (e \circ k \circ 0)(e \circ t \circ 0) = \theta(k)\theta(t) \end{aligned}$$

— indeed an embedding of the kernel into the constants. $\square$

Definition 3.3. Let $\langle R, +, \cdot, \circ \rangle$ be a triangular ring and let d be a differentiation of R . We call an exponential e of a differentiation d **multiplicative** if for all $f, g \in R$

$$e \circ (f + g) = (e \circ f) \cdot (e \circ g).$$

Definition 3.4. Let $\langle R, \cdot, \circ \rangle$ be a right ring and let C be the ideal of its constants. We call R **constant-extensional** or **C-extensional** if for any $f, g \in R$,

$$\forall c \in C : f \circ c = g \circ c \implies f = g.$$

Remark. The converse implication holds automatically by the definition of composition as an operation.

Definition 3.5. Let $\langle R, +, \cdot, \circ \rangle$ be a triangular ring. We call R **C-extensional** if $\langle R, \cdot, \circ \rangle$ is C -extensional.

4 Theorem 2. On the exponential of a linear differentiation

Let $\langle R, +, \cdot, \circ \rangle$ be a C-extensional triangular ring. Let C be the ideal of its constants and let d be a linear differentiation which is not identically zero and such that there exists $c \in C$ with $d(c) = 0$. Let E be the set of all exponentials of d and let E is not empty. Then the following are equivalent:

- (i) d has a multiplicative exponential e
- (ii) for any $e_1, e_2 \in E$, $e_1/e_2 \in C$

Remark. Condition (ii) is equivalent to E being a subset of a one-dimensional linear subspace of R over C ; in other words, all exponentials lie on a single ray over C .

Lemma 4.1. If a differentiation d has a constant c such that $d(c) = 0$, then for any constant $t \in C$, $d(t) = 0$:

$$d(t) = d(t \circ c) = d(t) \circ c \cdot d(c) = 0 \quad \square$$

Lemma 4.2. $\ker(d) = \{\text{id} + c \mid c \in C\}$.

Proof. Let $c \in C$. By linearity and Lemmas 4.1 and 2.5 (id is the identity element of the composition subgroup $\ker(d)$):

$$d(\text{id} + c) = d(\text{id}) + d(c) = 1 + 0 = 1,$$

so $\text{id} + c \in \ker(d)$.

Conversely, let $f \in \ker(d)$, that is, $d(f) = 1$.

By Lemma 2.5 and linearity,

$$d(f) = 1 = d(\text{id}) \implies d(f - \text{id}) = 0.$$

Since d is not identically zero, by Lemma 2.4, $f - \text{id} \in C$.

Hence $f = \text{id} + c$ for some constant $c \in C$. $\square$

Lemma 4.3. Let e be a multiplicative exponential of d . Then for any $c \in C$, $e \circ c \neq 0$.

Proof. Suppose there exists $c \in C$ such that $e \circ c = 0$.

Then, on the one hand,

$$e \circ 0 = e \circ (0 + 0) = (e \circ 0) \cdot (e \circ 0),$$

hence $e \circ 0 = 1$. On the other hand,

$$e \circ 0 = e \circ (c - c) = (e \circ c) \cdot (e \circ [-c]) = 0 \cdot (e \circ [-c]) = 0.$$

This gives a contradiction $0 = 1$. $\square$

Lemma 4.4. We show that (i) implies (ii).

Proof. Let e be a multiplicative exponential of d and let $e_1, e_2 \in E$. By the bijection theorem and Lemma 4.2, there exist $c_1, c_2 \in C$ such that

$$e_1 = e \circ (\text{id} + c_1), \quad e_2 = e \circ (\text{id} + c_2).$$

By the multiplicativity of e ,

$$e_1 = (e \circ \text{id}) \cdot (e \circ c_1), \quad e_2 = (e \circ \text{id}) \cdot (e \circ c_2).$$

In Lemma 3.10 of [1], we showed that the constants form a two-sided ideal in $\langle R, \cdot, \circ \rangle$; hence $e \circ c_1, e \circ c_2 \in C$. On the other hand, $e \circ c_1, e \circ c_2 \neq 0$ by Lemma 4.3, so they are invertible, and we can express $e \circ \text{id}$ from both sides and equate:

$$\frac{e_1}{e \circ c_1} = \frac{e_2}{e \circ c_2} \implies \frac{e_1}{e_2} = \frac{e \circ c_1}{e \circ c_2} \in C,$$

- since the constants form a subfield of $\langle R, +, \cdot \rangle$ by Lemma 2.1. $\square$

Lemma 4.5. Let e be an exponential of d and $\forall e_1, e_2 \in E: e_1/e_2 \in C$. Then $E = \{ce \mid c \in C \setminus \{0\}\}$.

Proof. Clearly, $E \subseteq \{ce \mid c \in C\}$, since for any $e' \in E$, by condition of the lemma,

$$e'/e \in C \quad \text{i.e.} \quad \exists c \in C: e' = ce$$

Let us show that $\{ce \mid c \in C \setminus \{0\}\} \subseteq E$. Let $c \in C$. Then, by the linearity of d ,

$$d(ce) = c d(e) = ce,$$

so $ce \in E$. We exclude 0, since it is constant and therefore not an exponential by definition. The remaining elements are not constant since $ce = c'$ implies $e = c'/c \in C$, as C is a field by Lemma 2.1. $\square$

Lemma 4.6. Let e' be an exponential of d and $\forall e_1, e_2 \in E: e_1/e_2 \in C$. Let us show that d has an exponential e that is multiplicative on C .

Proof. Since R is C -extensional, there exists $\gamma \in C$ such that $e' \circ \gamma \neq 0$; otherwise $e' = 0$ and would not be an exponential by definition. By the bijection theorem and Lemma 4.2, $e_\gamma = e' \circ (\text{id} + \gamma)$ is an exponential and

$$e_\gamma \circ 0 = e' \circ (\text{id} \circ 0 + \gamma \circ 0) = e' \circ (0 + \gamma) \neq 0.$$

Let $e = e_\gamma / (e_\gamma \circ 0)$. Note that $e_\gamma \circ 0$ is a nonzero constant, since the constants form a two-sided ideal by Lemma 3.10 of [1]; hence e is an exponential by Lemma 4.5, and

$$e \circ 0 = (e_\gamma / (e_\gamma \circ 0)) \circ 0 = (e_\gamma \circ 0) / (e_\gamma \circ 0) = 1.$$

Since $e \circ 0 = 1$ and d is a **derivative** by the hypothesis of the lemma, by Lemma 3.2, e defines an embedding θ of the kernel into the constants. Let $\text{id} + a, \text{id} + c \in \ker(d)$. Then, by Lemma 3.2:

$$e \circ [\text{id} + a] \circ [\text{id} + c] \circ 0 = (e \circ [\text{id} + a] \circ 0)(e \circ [\text{id} + c] \circ 0).$$

Expanding by the distributivity of composition over addition, and noting that a, c are constants, we obtain

$$e \circ (c + a) = (e \circ a)(e \circ c). \quad \square$$

Thus, we have shown that d has an exponential e that is multiplicative on C . To complete the proof of the theorem, it remains to extend its multiplicativity property to all of R . This follows from the C -extensionality of R .

Lemma 4.7. Let e' be an exponential of d and $\forall e_1, e_2 \in E: e_1/e_2 \in C$. We show that d has a multiplicative exponential e ; that is, (ii) $\implies$ (i).

Proof. In Lemma 4.6 we showed that d has an exponential e that is multiplicative on constants. Using this, we show that for any $c \in C$,

$$e \circ (f + g) \circ c = [(e \circ f) \cdot (e \circ g)] \circ c$$

Let $f, g \in R$ and let $c \in C$. **Then:**

$$e \circ (f + g) \circ c = e \circ (f \circ c + g \circ c)$$

Since $f \circ c, g \circ c \in C$ (as C is an ideal), we get:

$$= (e \circ f \circ c) \cdot (e \circ g \circ c) = [(e \circ f) \cdot (e \circ g)] \circ c$$

— by distributivity. Hence, by C -extensionality,

$$e \circ (f + g) = (e \circ f) \cdot (e \circ g)$$

This completes the proof of the theorem. $\blacksquare$

Corollary of Theorem 2. By Theorem 2 and Theorem 6 of [1], the following statements are equivalent for a linear differentiation that has an exponential and a zero at some point:

- all exponentials of d lie on a single ray over C ;
- d has a multiplicative exponential;
- d satisfies the Leibniz rule.

This indicates a critical error in the remark following Theorem 3 of [1] — our classification of linear differentiations over real functions was wrong. Clearly, the exponentials of the linear operators $d(x^n) = n^k x^{n-1}$ given in that paper lie on a single ray by linearity. This means that if these operators were indeed differentiations, then each of them would have a multiplicative exponential, and they would themselves satisfy the Leibniz rule. In turn, the Leibniz rule uniquely determines a differentiation on the basis by induction as $d(x^n) = nx^{n-1} \cdot d(x) = nx^{n-1}$. Therefore, the operators from [1] are not differentiations for $k \neq 1$.

However, we have decided not to correct the paper “On Algebraic Integral” and to leave everything as it is — the hypothesis on the existence of many linear differentiations seems too beautiful to simply delete it. Moreover, one cannot rule out that there exist worlds in which this hypothesis turns out to be true after all (who knows what universes with Heyting logic may delight us with), and perhaps one day we will be able to pass this knowledge on to them, unless Academia finally destroys science and civilization in this world before then.

There is nothing wrong with mistakes. For example, the curious identity in this paper was obtained as a result of my failing to solve an elementary differential equation for several days. And if humanity ever again has a system that truly engages in education rather than solving various political problems, I think that in such a system mistakes will be encouraged, and not punished at all.

5 Some differentiations of triangular rrings

The correct classification of differentiations over $C^1(\mathbb{R})$ was given by S. Artstein-Avidan, H. König, and V. Milman in the paper “The chain rule as a functional equation” [3]. By the classification of solutions to the functional equation

$$d(f \circ g) = d(f) \circ g \cdot d(g),$$

where $d : C^1(\mathbb{R}) \rightarrow C(\mathbb{R})$, every such differentiation is of the form:

$$d(f) = \frac{H \circ f}{H} |f'|^p, \quad p \geq 0,$$

where H is a positive continuous function. Moreover, for $p > 0$ there is an additional family of solutions given by

$$d(f) = \frac{H \circ f}{H} |f'|^p \operatorname{sgn}(f').$$

However, their classification, while likely correct, is substantially incomplete from an algebraic point of view. The authors explicitly use the fact that the domain of the differentiable functions is all of $\mathbb{R}$. At the same time, it is easy to see that the number of such functions forming a triangular rring is rather small and, strictly speaking, not very suitable for calculus — for example, the sine function is compositionally invertible only on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and we cannot use the arcsine in differential and integral computations if we restrict ourselves in this way.

Definition 5.1. Let $\langle R, +, \cdot, \circ \rangle$ be a C -extensional triangular rring and let C be the ideal of its constants. We call a linear differentiation of R a **derivative** if it has a zero at some point and a nonempty set of exponentials such that all its exponentials lie on a single ray over C . We denote such a differentiation by $d(f) = f'$.

Lemma 5.2. Let $\langle R, \cdot, \circ \rangle$ be a right ring and let C be the ideal of its constants. Then a differentiation d of R has no zeros **iff** it is of the form:

$$d(f) = \frac{H}{H \circ f}$$

— where $H \in (R \setminus \{0\})$.

Proof.

On the one hand, by Lemma 2.4, if a differentiation has a zero at a nonconstant point, it is identically zero, and we exclude it from consideration. On the other hand, by Lemma 4.1, if a differentiation is not equal to zero on all constants, then it is not equal to zero at any constant. Let d be a differentiation and let $c \in C$. Then

$$d(c) = d(c \circ f) = d(c) \circ f \cdot d(f)$$

Since $d(c)$ is nonzero, it is multiplicatively invertible, and hence $d(f)$ is uniquely determined by the value $d(c) = H$. Thus,

$$d(f) = \frac{H}{H \circ f}$$

Conversely, let d be of this form. Then for any $f, g \in R$,

$$\begin{aligned} d(f) \circ g \cdot d(g) &= \left(\frac{H}{H \circ f} \circ g \right) \cdot \frac{H}{H \circ g} \\ &= \frac{H \circ g}{H \circ f \circ g} \cdot \frac{H}{H \circ g} = \frac{H}{H \circ f \circ g} = d(f \circ g) \end{aligned}$$

Thus, d is indeed a differentiation. Assuming that d has a zero at some constant c leads to a contradiction:

$$\frac{H}{H \circ c} = 0 \implies H = 0,$$

and $d = 0/0 \square$

Lemma 5.3. Let R be a right ring. Then its differentiations form a multiplicative group.

Proof. Let $f, g \in R$ and let d, p be differentiations of R . We show that $q = d/p$ is also a differentiation:

$$\begin{aligned} q(f \circ g) &= \frac{d(f \circ g)}{p(f \circ g)} = \frac{d(f) \circ g \cdot d(g)}{p(f) \circ g \cdot p(g)} \\ &= \left(\frac{d(f)}{p(f)} \circ g \right) \cdot \frac{d(g)}{p(g)} = q(f) \circ g \cdot q(g) \quad \square \end{aligned}$$

Lemma 5.4. Let $\langle R, +, \cdot, \circ \rangle$ be a C-extensional triangular rring and let $' : R \rightarrow R$ be a derivative. Then operators of the following form are differentiations:

$$d(f) = \frac{H}{H \circ f} \cdot (f')^z$$

for $H \in (R \setminus \{0\})$, $z \in \mathbb{Z}$.

Proof. follows directly from Lemmas 5.2 and 5.3 $\square$

Remark. It is easy to see that all operators of the form $(H/H \circ f)^z$, $z \in \mathbb{Z}$, cannot have zeros if $\langle R, +, \cdot \rangle$ is an integral domain, so they reduce to $\tilde{H}/\tilde{H} \circ f$ by Lemma 5.2.

Remark. Note that the classification in [3] does not account for differentiations with $p < 0$, due to the domain of the operators considered there.

Remark. The complete classification of differentiations of triangular rings remains a subject of further research for our community.

The classification of exponentials and kernels of the differentiations from Lemma 5.4 is nontrivial. If the equation

$$\frac{H}{H \circ f} \cdot (f')^z = \frac{1}{\text{id}}$$

is solvable for all H, z , the only differentiation of this form that has no exponential is $d(f) = f/x$.

Lemma 5.5. Let the equation

$$\frac{H}{H \circ f} \cdot (f')^z = \frac{1}{\text{id}}$$

have a solution f for every choice of H and z . Let d be a differentiation of the form

$$d(f) = \frac{H}{H \circ f} \cdot (f')^z$$

and suppose d has no exponential. Then $d(f) = f/x$.

Proof.

By Theorem 4 of [1], a differentiation has no exponential if and only if the preimage of the element $1/\text{id}$ is empty or consists of constants. By the assumption of the lemma, we suppose it is never empty, hence, d has no exponential only if there exists a constant c such that

$$d(c) = \frac{H}{H \circ c} \cdot (c')^z = \frac{1}{\text{id}}.$$

Since the derivative vanishes on constants, the equation has a solution H only for $z = 0$ and reduces to

$$\frac{H}{H \circ c} = \frac{1}{\text{id}}.$$

$$H = \frac{H \circ c}{\text{id}}.$$

$H \circ c$ is a constant, since C is an ideal. Denoting it by t , we get $H = t/\text{id}$, and

$$d(f) = \frac{t/\text{id}}{(t/\text{id}) \circ f} = \frac{t/\text{id}}{t/f} = f/\text{id}. \quad \square$$

Remark. It is easy to verify that this differentiation is also linear, on a par with the derivative. Of particular interest are differentiations of the form

$$d(f) = \left(\frac{f}{\text{id}} \right)^m \cdot (f')^z, \quad m, z \in \mathbb{Z}.$$

That they are indeed differentiations follows immediately from the distributivity of composition over multiplication, by the converse part of Theorem 2 of [1] and Lemma 5.3. These differentiations correspond to $H = \text{id}^m$:

$$\frac{\text{id}^m}{\text{id}^m \circ f} = \frac{\text{id}^m}{f^m} = \left(\frac{f}{\text{id}} \right)^{-m}.$$

The classification of exponentials and kernels of such differentiations is not a difficult task. However, it requires a careful introduction of the notion of an antiderivative, which lies beyond the scope of this paper. Moreover,

such differentiations satisfy the functional identity discussed in the remark following Theorem 3 of [1]:

$$\theta(pq)x^{\psi(pq)} = \theta(p)x^{\psi(p)q} \cdot \theta(q)x^{\psi(q)}.$$

— where ψ is a crossed-homomorphism and φ is a multiplicative homomorphism. Operators of the form

$$d(f) = \left(\frac{f}{\text{id}} \right)^m \cdot (f')^z$$

are the only solution we have been able to find so far:

$$\begin{aligned} d(x^q) &= \left(\frac{x^q}{x} \right)^m \cdot ([x^q]')^z = x^{m(q-1)} \cdot (qx^{q-1})^z \\ &= q^z x^{m(q-1)+z(q-1)} = q^z x^{(m+z)(q-1)} \end{aligned}$$

However, it seems likely that among them only the derivative and f/id are linear. We cannot assert the uniqueness of such a solution, since all we have been able to prove so far about the continuation of differentiations is Theorem 3 of [1], as well as the following lemma.

Lemma 5.6. On the extension of a differentiation within an injective subgroup

Let $\langle R, \cdot, \circ \rangle$ be a right ring and let D be a subgroup of its differentiations such that every $d \neq 1$ in D is injective. Then, any differentiation in D is uniquely determined by its value at a single element $f \in (R \setminus \{\text{id}\})$. That is,

$$\forall d, p \in D, \quad \exists f \in (R \setminus \{\text{id}\}) : d(f) = p(f) \implies d = p.$$

Proof. Let $d, p \in D$ and let $f \in R \setminus \{\text{id}\}$ be such that $d(f) = p(f)$. Let $q = d/p \in D$. Then $q(f) = 1$. However, by Lemma 2.5, $q(x) = 1 = q(f)$. Hence q is not injective, so $q = 1$ and $d = p$. $\square$

Example. For example, it is easy to verify that differentiations having no zeros form a subgroup if $\langle R, +, \cdot \rangle$ is an integral domain. On the other hand, from Lemma 5.2 it is easy to see that the kernels of such differentiations are trivial:

$$\frac{H}{H \circ f} = 1 \implies H = H \circ f$$

that is, the only solution is $f = \text{id}$. Hence, by the corollary of Lemma 2.6, all of them except the identity 1 are injective and are uniquely determined by their value at a single point.

6 Conclusion

Differentiations are truly beautiful.

Example 1. For example, we see that the differentiation

$$d(f) = \left(\frac{f}{\text{id}} \right)^{-1} f'$$

maps the basis x^n to $\mathbb{N}$:

$$\left(\frac{x^n}{x} \right)^{-1} (x^n)' = x^{1-n} \cdot nx^{n-1} = n,$$

that is, it is constant on the basis, and hence is a multiplicative homomorphism on it:

$$d(x^n \circ x^m) = d(x^n) \circ x^m \cdot d(x^m) = d(x^n) \cdot d(x^m).$$

Example 2. The differentiation

$$d(f) = \left(\frac{f}{\text{id}} \right) f'$$

satisfies:

$$d(fg) = d(f)g^2 + f^2d(g).$$

By the corollary of Theorem 2, the derivative satisfies the Leibniz rule; therefore,

$$\begin{aligned} d(fg) &= \frac{fg(fg)'}{\text{id}} = \frac{fg(f'g + fg')}{\text{id}} \\ &= \frac{ff'g^2}{\text{id}} + \frac{f^2gg'}{\text{id}} = d(f)g^2 + f^2d(g) \quad \square \end{aligned}$$

Curious identity. Another differentiation

$$d(f) = \left(\frac{f}{\text{id}} \right)^{-1} (f')^2$$

satisfies the identity:

$$d(f+g) = \frac{d(fg) + d(f)(f-g) + d(g)(g-f)}{f+g}$$

Proof.

On the one hand, by the Leibniz rule,

$$\begin{aligned} d(fg) &= \frac{\text{id} \cdot ((fg)')^2}{fg} = \frac{\text{id} \cdot (f'g + fg')^2}{fg} \\ &= \frac{\text{id} \cdot ((f'g)^2 + 2f'gfg' + (fg')^2)}{fg} \\ &= \frac{\text{id} \cdot (f')^2g}{f} + \text{id} \cdot 2f'g' + \frac{\text{id} \cdot (g')^2f}{g} \\ &= d(f)g + \text{id} \cdot 2f'g' + fd(g) \end{aligned}$$

so that

$$\text{id} \cdot 2f'g' = d(fg) - d(f)g - fd(g)$$

On the other hand,

$$d(f + g) = \frac{\text{id} \cdot ((f + g)')^2}{f + g}$$

that is, by the linearity of the derivative,

$$\begin{aligned} d(f + g)(f + g) &= \text{id} \cdot (f' + g')^2 \\ &= \text{id} \cdot (f')^2 + \text{id} \cdot 2f'g' + \text{id} \cdot (g')^2 \\ &= d(f)f + \text{id} \cdot 2f'g' + d(g)g \end{aligned}$$

so that

$$\text{id} \cdot 2f'g' = d(f + g)(f + g) - d(f)f - d(g)g$$

Equating the two expressions, we obtain

$$d(f + g)(f + g) - d(f)f - d(g)g = d(fg) - d(f)g - fd(g)$$

$$d(f + g)(f + g) = d(f)f - d(f)g + d(g)g - d(g)f \quad \square$$

These examples prove something far greater than any theorem — that mathematics is first and foremost the science of the beauty of the world in all its diversity, and the closest sister of aesthetics, and not at all a screwdriver for solving physical problems, as the modern Academy tries to present it. However, we shall not despair, knowing that sooner or later such attempts will fail, for nature does not tolerate violence against itself.

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